PARMCO - Checkpoint 1

AI-generated assembly instructions and actual reported parts list | Updated October 19, 2026

Current status: The motor stand and both bypass capacitors are installed. The four SN754410NE input pulldown resistors are 1.8 kohm each. The lab partner reports the motor worked during physical testing. Electrical stepping signals for both commanded directions were verified with a multimeter; actual physical direction and exact one-revolution behavior are not independently documented.

Parts list

Component	Qty	Use / status
Raspberry Pi 4 Model B Rev 1.2	1	Controller; BCM GPIO
SN754410NE DIP-16	1	Dual H-bridge
4N25 DIP-6	1	Optocoupled enable
NEMA17 XY42STH34-0354A	1	Motor; tested per partner
External 5 V and 12 V bench supply	1	Logic and motor rails
330 ohm resistor (R1)	1	GPIO24 to 4N25 LED
10 kohm resistor (R2)	1	Enable net pulldown
1.8 kohm resistors (R3-R6)	4	SN input pulldowns; installed
100 nF capacitors (C1-C2)	2	Supply bypass; installed
Breadboard and jumper wires	As used	Interconnections
Motor stand	1	Installed
Propeller	As used	Stall-speed test; status confirm
Digital multimeter, safety eyewear	As used	Verification and safety

Assembly instructions

1. With all supplies off and RP4 unplugged, orient the ICs using their package notches and verify pin numbering against the datasheets.

2. Wire SN pin 16 to external 5 V (VCC1) and pin 8 to external 12 V (VCC2). Connect SN pins 4, 5, 12, 13, RP4 GND and bench-supply ground together.

3. Wire RP4 GPIO24 (physical 18) through R1 (330 ohm) to 4N25 pin 1; 4N25 pin 2 to GND; pin 5 to external 5 V; pin 4 to SN pins 1 and 9. Install R2 (10 kohm) from this enable node to GND. Leave 4N25 pins 3 and 6 unconnected.

4. Connect RP4 GPIO17, GPIO27, GPIO22 and GPIO23 to SN input pins 2, 7, 10 and 15, respectively. Install one 1.8 kohm pulldown from EACH input pin to common ground (R3-R6).

5. Install C1 (100 nF) from SN VCC1 pin 16 to GND and C2 (100 nF) from SN VCC2 pin 8 to GND; both capacitors were reported installed.

6. With power off, confirm winding pairs by resistance. Green/grey (~37 ohm) connect to SN outputs pins 3/6; red/yellow (~36.3 ohm) connect to outputs 11/14. Secure motor on installed stand before powered operation.

Assembly instructions (continued)

7. Complete and record the student-written pre-power inspection and use instructor-directed current limits before motor-powered operation. The SN754410NE does not regulate winding current.

Verification and motor testing

Verified electrically: Commanded clockwise and counterclockwise full-step GPIO patterns matched the expected HIGH (~3.14–3.17 V) and LOW (~0 V) levels at driver inputs. ARM generic timer frequency: 54,000,000 Hz. 4N25 enable HIGH: approximately 5 V.

Reported by lab partner: The motor operated during physical testing. The following stall-speed trial readings and observed erratic behavior were supplied by the lab partner; these are reported measurements, not independently re-measured by the document author.

Stall-speed characterization — full-step mode

Trial	Rate (steps/s)	Calculated RPM	Observation
1	338.64	101.59	Erratic behavior; slower than prior step
2	340.79	102.24	Erratic behavior
3	339.47	101.84	Erratic behavior

Average: 339.63 full-steps/s; **average RPM:** 101.89; **trial spread:** 0.63%.

Stall-speed characterization — half-step mode

Trial	Rate (steps/s)	Calculated RPM	Observation
1	551.60	82.74	Erratic behavior
2	572.68	85.90	Erratic behavior
3	573.39	86.01	Erratic behavior

Average: 565.89 half-steps/s; **average RPM:** 84.88; **trial spread:** 3.85%.

Calculation: RPM = (steps per second × 60) / steps per revolution; 200 full steps/rev or 400 half steps/rev. Spread = (maximum rate – minimum rate) / average rate × 100%.

Comparison to 200 RPM target: Both reported operating limits are below 200 RPM (full-step: ~101.89 RPM mean; half-step: ~84.88 RPM mean). Report these findings to the instructor as the assignment directs. The recorded erratic behavior suggests loss of reliable stepping at these tested rates; do not present them as confirmed successful stall-free speeds.

Safety and submission notes

Disconnect power before moving wires; secure the motor and wear eye protection when a propeller spins. Keep the separate student-written pre-power inspection based on actual observations. Verify all AI-generated details against the physical build before submission.